

# Compressed Traffic Assignment: Numerical Report v4.0

Certified results of the lockstep campaign (2026-07-25 – 2026-07-27)

Campaign record — `campaign.weighted/`, branch `version.3`

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**Headline, Chicago Sketch (path-rich regime).** On the nested-pool ladder with a fixed set of 19,112 OD pairs (94.2% of demand) and matched ALM operators, compression with a rank-6 backbone and at most 500 dynamically repriced OD exchange atoms attains

$$S(G^*=0.015) = \mathbf{1.87} \times \quad (\bar{K} = 31.6, n = 603,232), \quad S(G^*=0.015) = 1.46 \times \quad (\bar{K} = 25.4),$$

at *matched time-to-accuracy against a converging full-space baseline*. The advantage rises monotonically with route richness and with target tightness inside the reachable band; below  $\bar{K} \approx 15$  compression does not pay ( $S = 0.81$  at  $\bar{K} = 14.6$ ; does not reach the targets at  $\bar{K} = 7.7$ ).

**Headline, Chicago Regional (structural negative, explained).** The E0 pool has  $\bar{K} = 7.48$ , so the reduction ceiling (11) caps compression at  $1 - 1/7.48 = 86.6\%$  — exactly the measured saturation value ( $84.0 \rightarrow 86.3 \rightarrow 86.6 \rightarrow 86.6\%$  for  $\tau = 1.03, 3, 8, 20$ ; at  $\tau=20$  the split retains 270,716 majors against 270,598 OD pairs, i.e. one path per OD, the hard floor). The compiled solver paid only above  $\sim 90\%$  reduction, hence Regional *cannot* pay with this pool for any  $(\tau, r)$ : the certified C++ bound is  $S < 0.58 \times$ . The negative is a property of the pool’s richness, not of the method or its tuning, and is predictable in advance from two integers ( $n$  and  $\ell$ ).

[Note to co-authors: Everything in this report traces to CSVs in `campaign.weighted/results/` (commits `f6deb4e`, `4612cae`, `a25b5a4`, `a01413d`, `165e816`). All timings single-process, single-thread. The old “ $3.26 \times$  with essentially zero accuracy loss” phrasing is withdrawn; see §6.]

## 1 Model, algorithms, and metrics

**Assignment model.** Path flows  $x \in \mathbb{R}_{\geq 0}^n$  over a pool of  $n$  paths and  $\ell$  OD pairs, incidence  $B \in \{0, 1\}^{n \times m}$ , OD aggregation  $A \in \{0, 1\}^{\ell \times n}$ , demand  $d \in \mathbb{R}^\ell$ , BPR link costs  $t(v)$ ,  $v = v_0 + B^\top x$ . Beckmann objective

$$Z(x) = \sum_{a=1}^m \int_0^{v_a} t_a(s) ds. \quad (1)$$

**Operator (matched on both sides).** SPG-ALM: at outer iteration  $k$ , minimise over the representation’s variables

$$\mathcal{L}_\rho(x, \lambda) = Z(x) + \lambda^\top (Ax - d) + \frac{\rho}{2} \|Ax - d\|_2^2, \quad \lambda \leftarrow \lambda + \rho (Ax - d), \quad (2)$$

inner solver L-BFGS-B (maxiter 200, gtol  $10^{-5}$ ),  $\rho_0 = 100$  doubled when the constraint residual fails to contract by  $4 \times$ , identical schedules in every arm.

**Compressed representation (incumbent).** Major/minor split at threshold  $\tau$  (every OD keeps its maximum-flow path explicit); minors are decoded through a global signed basis with an affine anchor:

$$x_{\mathcal{M}} = w_0 + U_r z, \quad \text{feasibility } w_0 + U_r z \geq 0 \text{ (ALM hinge, multiplier } \mu), \quad (3)$$

$U_r$  = leading left singular vectors of  $SB_2$ ,  $S = \text{diag}(\sqrt{w_0} + \varepsilon)$  (flow-weighted *row-scaling heuristic*; it is *not* the minimiser of the flow-weighted reconstruction loss, which is  $\text{span}(S^{-1}\tilde{U}_r)$  — verified 2026-07-25 and stated wherever the basis is described).

**Hybrid global–local representation (new).** A low-rank backbone plus sparse, OD-local, *coefficient-bounded* route-exchange atoms on an active set  $\mathcal{A}$ :

$$x_{\mathcal{M}} = w_0 + U_G z_G + \sum_{q \in \mathcal{A}} \alpha_q a_q, \quad a_q = \frac{1}{\sqrt{2}}(e_{p_q^{\text{low}}} - e_{p_q^{\text{high}}}), \quad \mathbf{1}^\top a_q = 0, \quad (4)$$

where  $p_q^{\text{low}}/p_q^{\text{high}}$  are the cheapest/costliest *used* minors ( $x > \varepsilon$ ) of OD  $q$  at current costs, and

$$-\sqrt{2}w_0[p_q^{\text{low}}] \leq \alpha_q \leq \sqrt{2}w_0[p_q^{\text{high}}] \quad (5)$$

so the atom part alone cannot violate nonnegativity: atoms encode *flow exchange*, not signed reconstruction. The active set is chosen by per-OD disequilibrium

$$G_q = \sum_{p \in P_q} x_p (c_p - c_q^{\min}), \quad \mathcal{A} = \{q : G_q \text{ among the top } B\}, \quad (6)$$

and, in the *dynamic* variant, the anchor  $w_0$ , the costs,  $G_q$  and the atom pairs are re-computed every outer iteration (equilibrium-error-guided column generation in the tangent space);  $\lambda, \rho$  carry across repricings.

**Independent evaluation (the third evaluator).** Every claim uses one independently implemented full-space scorer, applied to the raw decoded flows of *both* solvers — never a solver’s internal gap:

$$G(x) = \frac{\sum_q \sum_{p \in P_q} x_p (c_p - c_q^{\min})}{\sum_q d_q c_q^{\min}} \quad (\text{common equilibrium gap, pool-min costs}), \quad (7)$$

$$r_{\text{OD}} = \frac{\sum_q |\sum_{p \in P_q} x_p - d_q|}{\sum_q d_q}, \quad r_{\text{OD}, \max} = \max_q \frac{|\sum_p x_p - d_q|}{\max(d_q, \epsilon)}, \quad (8)$$

$$r_{\text{PG}} = \frac{\|x - \Pi_{\mathcal{X}}(x - \alpha \nabla Z(x))\|_2}{\max(1, \|x\|_2)}, \quad \alpha = 10^{-2} \quad (\Pi_{\mathcal{X}}: \text{per-OD Euclidean simplex proj.}). \quad (9)$$

Speed claims are *time-to-target*

$$S(G^*) = \frac{T_{\text{full}}(G^*)}{T_{\text{comp}}(G^*)}, \quad (10)$$

never final wall time after a solver has stalled. Repeated-scenario economics use  $H^* = \lceil T_{\text{prep}} / (T_{\text{warm}} - T_{\text{comp}}) \rceil$ .

## 2 Result 1: the reduction ceiling

Every OD pair must keep at least one explicit path, so route richness  $\bar{K} = n/\ell$  caps the dimensional reduction *before any tuning*:

$$\text{reduction} \leq \frac{n - \ell - r}{n} \approx 1 - \frac{1}{\bar{K}}. \quad (11)$$

Table 1: The ceiling binds everywhere (exp06; `exp06_regional_tau.csv`).

| instance                     | $\bar{K}$ | ceiling $1-1/\bar{K}$ | measured reduction | note                                     |
|------------------------------|-----------|-----------------------|--------------------|------------------------------------------|
| Regional-E0 ( $\tau=1.03$ )  | 7.48      | 86.6%                 | 84.0%              |                                          |
| Regional-E0 ( $\tau=3$ )     | 7.48      | 86.6%                 | 86.3%              |                                          |
| Regional-E0 ( $\tau=8, 20$ ) | 7.48      | 86.6%                 | <b>86.6%</b>       | saturated: 270,716 majors vs 270,598 ODs |
| Sketch-K10                   | 11.09     | 91.0%                 | 89.8%              |                                          |
| Sketch-K15                   | 15.34     | 93.5%                 | 92.4%              |                                          |

The compiled solver paid only above  $\sim 90\%$  reduction; Regional-E0 cannot reach it, hence the certified bound  $S < 0.58 \times$  (full C++ solve 1085.3s, objective 18,854,545.32 reproduced bit-for-bit; compressed stopped at  $>1860$ s). The old Regional negative is thereby *explained and predictable*: a network needs  $\bar{K} \gtrsim 10\text{--}15$  before compiled compression can be expected to pay.

## 3 Result 2: the representation floor

**Instrument.** Full and compressed ALM advance *one outer iteration at a time* from a byte-identical iteration 0 ( $w_0 = x_{\mathcal{M}}^0$ ,  $z^0 = 0$ ,  $y^0 = x_{\text{maj}}^0$ ; identity gate: all differences exactly 0.0), scored each step by (7)–(10) (exp08, Chicago Sketch K10, scenario d05 = demand  $\times (1 + U(-0.05, 0.05))$ , seed 100,  $r = 6$ ).

Table 2: Ten synchronized outer iterations (abridged; `exp08_lockstep_K10_d05_r6.csv`).

| $k$ | $G_{\text{full}}$ | $G_{\text{comp}}$ | $r_{\text{PG,full}}$ | $r_{\text{PG,comp}}$ | cpu <sub>full</sub> (s) | cpu <sub>comp</sub> (s) |
|-----|-------------------|-------------------|----------------------|----------------------|-------------------------|-------------------------|
| 1   | 0.00503           | 0.01563           | 0.00278              | 0.00275              | 41.5                    | 15.6                    |
| 4   | 0.00110           | 0.01358           | 0.00032              | 0.00105              | 37.5                    | 14.2                    |
| 7   | 0.00074           | 0.01370           | 0.00009              | 0.00103              | 39.6                    | 21.3                    |
| 10  | <b>0.00069</b>    | <b>0.01338</b>    | 0.00003              | 0.00102              | 50.8                    | 1.3                     |

OD feasibility is comparable throughout; the divergence is purely equilibrium quality. Kernel accounting: the full side spends 96% of CPU in gradient + link aggregation; the compressed side spends **74%** in the dense  $O(n_{\mathcal{M}}r)$  decode + hinge; the OD loop is 3–4% on both sides. A  $10\times$  variable reduction delivers only  $2.4\times$  per-outer cost.

**Five controls (exp09), branched from the same  $k=4$  state.**

- C1 *Shared duals* (both directions): full escapes under either side’s  $\lambda$  ( $G = 0.00091/0.00092$ ); compressed stays at its plateau (0.0132/0.0138). Multiplier drift eliminated.
- C2 *Hinge accounting*: at the plateau the iterate sits on the box boundary ( $\min u = -1.1 \times 10^{-2}$ ,  $H = 4.7 \times 10^3$ ,  $\|\nabla_z H\| = 2.3 \times 10^3$ ); removing the hinge yields  $G = 2.22$ . The hinge is *necessary but not the root cause*: the feasible set is  $\text{span}(U) \cap \{w_0 + Uz \geq 0\}$ .

- C3 *Gradient geometry*: with  $g_T$  = per-OD-centred path costs,  $\eta_{\text{captured}} = 0.134$ ,  $\eta_{\text{missing}} = 0.866$ ; 87.7% of  $\|g_T\|^2$  lies in the minor block, of which  $\text{span}(U_6)$  captures **1.26%** (measured at the  $r=6$  plateau; not extrapolated to other ranks).
- C4 *Full-space escape* from the identical iterate and ALM state:  $G$  0.0135  $\rightarrow$  0.00277 in *one* step, while a paired compressed step does nothing (0.0138). Decisive.
- C5 *Rank replay*: plateaus 0.0134/0.0135/0.0127 for  $r = 6/10/20$  — the limitation persists over the tested rank range.

$$G_{\min}(U, r) \approx 1.3 \times 10^{-2} \text{ for the tested global-SVD representation, } r \in \{6, 10, 20\} \quad (12)$$

*Replication*: on K15-d05 the plateau is 0.0162, the escape step gives  $0.01618 \rightarrow 0.00415$ , and  $\eta(\text{minor}) = 1.58\%$  — same mechanism, second pool. *Mechanism (exact wording)*: the global latent basis poorly aligns with the remaining OD-specific equilibrium directions, and the small aligned component is further restricted by the nonnegativity boundary of the decoded minor flows.

## 4 Result 3: structured representations and dynamic atoms

Table 3: One ALM step from the identical K10-d05 plateau point, same  $(\lambda, \rho)$  (exp10); then full trajectories from iteration 0 (exp10b/exp11).  $G_0 = 0.01366$ .

| representation                      | $R$                 | $\eta_{\text{minor}}$ | $G$ after 1 step | cpu (s) | trajectory plateau               |
|-------------------------------------|---------------------|-----------------------|------------------|---------|----------------------------------|
| global $r=6$ (incumbent)            | 6                   | 1.27%                 | 0.01094          | 28.5    | 0.01352                          |
| origin blocks, light                | 1,081               | 4.10%                 | 0.00819          | 28.5    | 0.01132                          |
| origin blocks, medium               | 2,792               | 9.07%                 | 0.00776          | 34.5    | 0.01123                          |
| odx1 (flow-ranked atoms)            | 69,707              | 5.06%                 | 0.01167          | 33.8    | —                                |
| odx1g (cost-selected, static)       | 69,707              | —                     | —                | —       | 0.01169 (converges cleanly)      |
| odxfull (within-OD zero-sum)        | 731,612             | <b>98.48%</b>         | 0.00583          | 156.3   | —                                |
| full-space escape (reference)       | 903,516             | 100%                  | 0.00277          | 97.3    | 0.00069 ( $k=10$ )               |
| static hybrid atoms, $B \leq 5000$  | (exp11)             |                       | —                | —       | 0.0103–0.0112                    |
| <b>dynamic repricing</b> , $B=2000$ | $\sim 1,300$ active |                       | —                | —       | <b>0.00407, still descending</b> |

Three certified statements. (i) *OD-locality contains essentially all the missing gradient*: the within-OD zero-sum minor subspace captures 98.48% (major–minor exchange is only  $\sim 1.5\%$ ): the floor is cross-OD coupling. (ii) *No cheap static locality removes the floor*: origin blocks plateau at  $\sim 0.0113$  regardless of budget; statically priced atoms at any  $B$  plateau at 0.0103–0.0112. (iii) *Dynamic repricing is the mechanism*: rebuilding cost-selected, coefficient-bounded atoms (4)–(6) each outer reaches  $G = 0.00407$  (still descending at the outer budget), passing  $10^{-2}$  at 40s and  $5 \times 10^{-3}$  at 103s, with the active set self-pruning ( $1,480 \rightarrow 1,158$ ) as ODs equilibrate. Stale atoms, not atoms, were weak. Exchange atoms also *converge cleanly* (inner iterations collapse to single digits; no boundary churn), unlike every signed-basis arm.

## 5 Result 4: the break-even transition (Chicago Sketch ladder)

Fixed top-20,000 ODs by demand (94.2% of total volume; 19,112 after singleton removal), nested penalty-KSP pools  $P_8 \subset P_{16} \subset P_{32} \subset P_{64}$  (dedupe; length filter  $c_p \leq 2.5 c_q^{SP}$  — the topology supports  $\bar{K}$  up to  $\sim 31$ ), matched ALM pairs, pre-registered  $G^* \in \{0.020, 0.015\}$  (exp12\_extreme\_M20000.csv).

Table 4: Time-to-target and speedup (10). Compressed arm: global  $r=6$  backbone +  $B \leq 500$  dynamically repriced atoms (repriced on stall).

| pool | $\bar{K}$ | $n$     | $T_{\text{full}}(.020)$ | $T_{\text{comp}}(.020)$ | $S(.020)$   | $T_{\text{full}}(.015)$ | $T_{\text{comp}}(.015)$ | $S(.015)$   |
|------|-----------|---------|-------------------------|-------------------------|-------------|-------------------------|-------------------------|-------------|
| X8   | 7.7       | 147,088 | 11.0 s                  | DNS                     | —           | 20.9 s                  | DNS                     | —           |
| X16  | 14.6      | 278,511 | 35.8 s                  | 44.0 s                  | 0.81        | 35.8 s                  | DNS                     | —           |
| X32  | 25.4      | 484,580 | 59.6 s                  | 44.7 s                  | <b>1.33</b> | 86.3 s                  | 59.3 s                  | <b>1.46</b> |
| X64  | 31.6      | 603,232 | 72.7 s                  | 57.0 s                  | <b>1.28</b> | 106.4 s                 | 57.0 s                  | <b>1.87</b> |

$S(G^*=0.015)$  is *monotone in richness* (DNS  $\rightarrow$  DNS  $\rightarrow$  1.46  $\rightarrow$  1.87): the advantage grows with the target inside the reachable band. Caveats, stated wherever these numbers appear: the  $\tau$  split leaves 25% majors (75% reduction, not the  $\sim 98\%$  extreme configuration — a tighter split is the obvious next lever and these  $S$  are likely understatements); Python engine; outer-granularity timing; one seed; controlled demand subset.

## 6 Requalified claims and retired claims

- **Say:** “On Chicago Sketch K15, the compiled compressed solver achieved  $3.26\times$  (wall time) with a 0.046% Beckmann-objective difference *in the tested moderate-equilibrium-accuracy regime*; the lockstep evaluator subsequently showed both sides of that comparison operated at common gap  $G \gtrsim 0.015$ .” **Do not say:** “ $3.26\times$  with essentially zero accuracy loss.”
- **Say:** “Chicago Regional cannot pay with the E0 pool:  $\bar{K} = 7.48$  caps reduction at 86.6% (11), below the  $\sim 90\%$  paying regime; certified bound  $S < 0.58\times$ .” **Do not** present Regional as a tuning failure.
- **Say:** “Compression pays above a measurable richness threshold ( $\bar{K} \approx 15\text{--}25$  here) at moderate accuracy targets, with  $S(0.015)$  up to  $1.87\times$ , and provably cannot pay below the richness ceiling or beneath the representation floor (12).”
- Repeated scenarios (grid, Python): compressed 14.1 s vs 50.1 s warm-started full ( $3.55\times$ ),  $H^* = 2$ ; on K10 in C++ the warm-started full solver reaches its own floor almost instantly, so reuse claims are engine- and instance-specific.
- The flow-weighted basis is a row-scaling heuristic (kept because it improves the solved feasible gap in 27/34 paired comparisons), not reconstruction-optimal.
- Timing noise floor  $1.45\times$ : no smaller wall-time ratio is claimed anywhere.

## Reproduction

```
# floor + controls          python drivers/exp08_lockstep.py --case K10 --scenario d05
                             python drivers/exp09_controls.py --case K10 --scenario d05
# structured representations  python drivers/exp10_structured.py ; exp10b_trajectory.py
# hybrid + dynamic repricing  python drivers/exp11_hybrid.py
# break-even ladder           python drivers/exp12_gen_pools.py --M 20000 --K 64 --ratio 2.5
                             python drivers/exp12_extreme.py --M 20000
```

Commits (branch version.3): f6deb4e, 4612cae, a25b5a4, a01413d, 165e816. Determinism caveat: ARPACK start vectors make  $U$  reproducible only up to column signs across processes; every controls driver therefore re-derives its branch state in-process.